



INTERNSHIP REPORT

POWER ELECTRONICS AND INDUCTION MOTOR CONTROL
V/f CONTROL AND FIELD-ORIENTED CONTROL IMPLEMENTATION

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Chapter 1

Internship Context and Objectives

1.1 Introduction

Electrical drives constitute an important application of power electronics and automatic control, combining electrical machines, power converters, measurement systems, and real-time control algorithms. The performance of a modern drive depends not only on the theoretical control strategy but also on its practical implementation, including signal acquisition, embedded computation, switching control, and experimental validation.

The internship presented in this report focused on the **experimental implementation of control techniques for induction motor drives**. The work was carried out using a real three-phase power converter, an induction motor, electrical measurement and sensing systems, and an STM32-based digital control platform.

The development followed a progressive approach, beginning with the preparation and characterization of the experimental system and continuing toward the implementation of increasingly advanced motor-control strategies. Particular attention was given to the transition from theoretical and simulation-based control algorithms to their implementation and validation on real hardware.

1.2 Host Laboratory and Technical Environment

The internship was conducted at **Microtech Lab, Boumerdès**, a laboratory working on experimental and technological developments in several engineering fields. Its activities cover areas including power electronics, electrical drives, control systems, renewable energy, electric vehicles, telecommunications, and computer engineering.

The internship was carried out within the **Power Electronics and Electrical Drives department**, where the experimental work focused on the control of electrical machines and the associated power electronic systems.

The laboratory environment provided access to the **equipment** required for the development and testing of electrical drive systems, including:

- Three-phase power electronic converters;
- Electrical machines and laboratory loads;
- Voltage and current measurement systems;
- Digital control hardware;
- STM32 microcontrollers;
- Oscilloscopes and laboratory instrumentation;
- Simulation and model-based development tools.

This environment made it possible to investigate the complete development chain of an electrical drive, from the mathematical formulation of the control algorithm to its real-time execution and experimental validation.

1.3 Internship Context

The internship lasted approximately **Three months** and was undertaken as part of an engineering curriculum in the field of power engineering, power electronics, and electrical drives.

The main purpose was to complement the theoretical study of electrical drive control techniques with their practical implementation on real hardware. The work therefore extended beyond numerical simulation and included the configuration of embedded hardware, generation of inverter switching signals, acquisition of electrical measurements, development of speed feedback, and experimental testing.

The initial work plan was based on a set of control techniques studied during the academic curriculum. These included scalar control, field-oriented control, direct torque control, and space vector modulation.

Because of the limited duration of the internship and the progressive nature of the experimental development, the different techniques were approached sequentially. This allowed each fundamental component of the drive system to be developed and validated before being incorporated into more advanced control structures.

1.4 Objectives and Technical Scope

The primary objective was to develop and experimentally validate control techniques for a three-phase induction motor drive.

The initial technical objectives included:

- Implementation of open-loop scalar control;
- Implementation of closed-loop scalar control;
- Implementation of Indirect Rotor Field-Oriented Control (IRFOC);
- Investigation of Direct Torque Control (DTC);
- Implementation and application of Space Vector Modulation (SVM).

To achieve these objectives, several preliminary tasks were necessary. These included the identification of the induction motor parameters, development of the inverter modulation stage, configuration of the STM32G4 control platform, establishment of communication with the development environment, and development of a speed measurement system.

The implementation process therefore covered several interconnected technical areas:

Electrical machine → Power converter → Measurement system → Digital controller → Control algorithm → Modulation → Experimental validation

The work ultimately resulted in the experimental implementation of **scalar control, Indirect Rotor Field-Oriented Control, and Space Vector Modulation on real hardware.**

The development of the system also involved model-based design using **PLECS and embedded code generation**, alongside direct embedded development using

STM32CubeIDE. This provided an opportunity to compare and combine simulation-based development with low-level microcontroller implementation.

An additional part of the work consisted of developing a **low-cost optical speed** measurement system. An infrared sensor and a custom rotating disk were used to generate the speed feedback signal required for closed-loop operation.

Consequently, the internship covered not only the control algorithms themselves but also the practical hardware and software infrastructure necessary for their implementation.

1.5 Report Organization

The report is organized according to the progressive development of the electrical drive system.

Chapter 2 presents the experimental platform used throughout the internship. It describes the induction motor, its electrical parameters, the power converter, measurement and sensing systems, and the STM32-based control architecture. The preliminary implementation of PWM, SPWM, and SVM is also presented, together with the model-based development environment and experimental validation of the inverter control signals.

Chapter 3 focuses on the implementation of **scalar control** of the induction motor. The theoretical operating principle is first introduced, followed by the development of the open-loop and closed-loop structures and their experimental evaluation.

Chapter 4 presents the development of **Indirect Rotor Field-Oriented Control**. The induction motor model, coordinate transformations, rotor-flux orientation, slip calculation, control structure, modulation stage, embedded implementation, and experimental results are discussed.

Chapter 5 concludes the report by summarizing the main achievements of the internship, the skills developed through the experimental work, and possible improvements and future developments of the drive system.

CHAPTER 2

EXPERIMENTAL PLATFORM AND PRELIMINARY DEVELOPMENT

2.1 Introduction

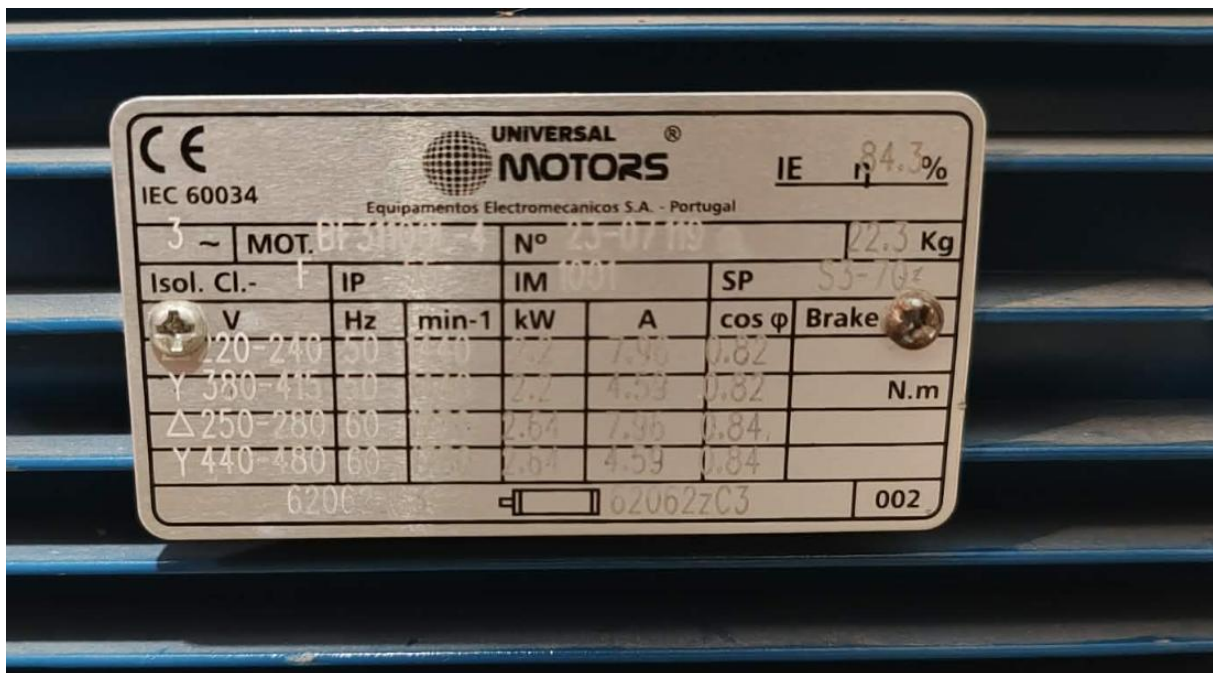
The experimental work was carried out on a three-phase induction motor drive platform developed in the Power Electronics and Electrical Drives laboratory of Microtech Lab. The platform combines a three-phase inverter, electrical measurement system, STM32-based digital control, and PLECS simulation and code-generation tools.

The preliminary stage consisted of identifying the motor parameters, commissioning the inverter, validating the measurement system, and establishing the communication between the

simulation environment and the embedded controller. These steps provided the necessary foundation for the implementation of the motor-control strategies.

2.2 Induction Motor and Parameter Identification

The machine used for the experiments was a three-phase induction motor manufactured by Universal Motors, Portugal. The motor nameplate supports several voltage and frequency configurations. The main nameplate information was recorded before the experimental tests.



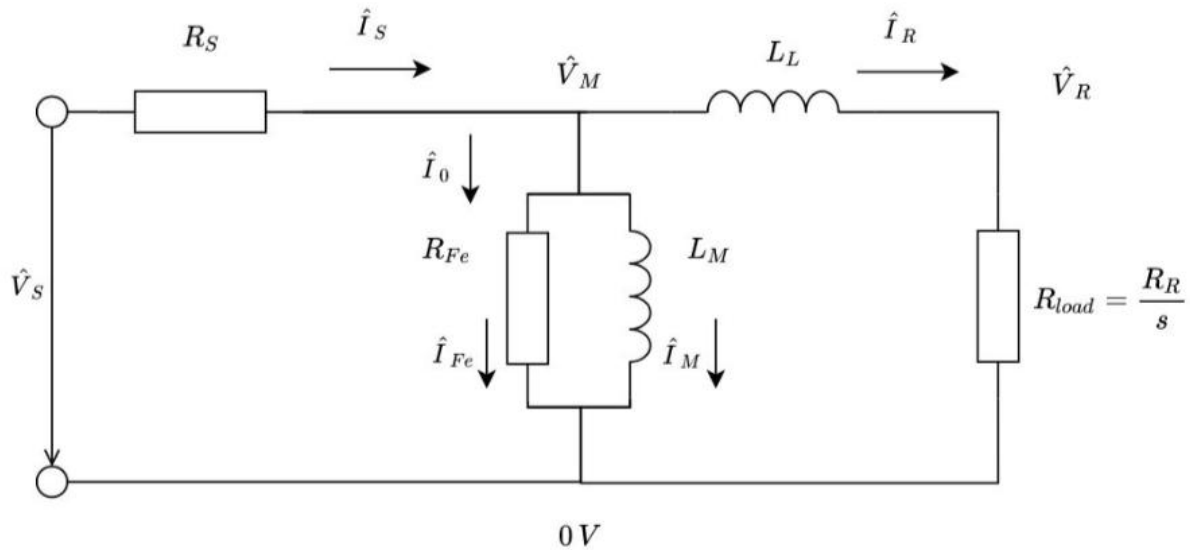
The electrical parameters required for the motor model were obtained experimentally using DC resistance, no-load, and locked-rotor tests.

Stator Resistance The resistance between two stator terminals was measured as:

$R_{LL} = 4.6 \text{ ohm}$ The stator phase resistance was therefore calculated as:

$$R_s = R_{LL} / 2 = 2.3 \text{ ohm}$$

No-Load Test During the no-load test, the motor was operated without an external mechanical load. The measured active power was used to separate the stator copper losses from the iron losses. The stator losses were calculated using:



EQUIVALENT CIRCUIT FOR INDUCTION MOTOR

$$P_{\text{stator}} = 3 \times R_s \times I^2$$

The iron-loss resistance was then obtained from the equivalent circuit. The identified value was:

$$R_f = 1090.51 \text{ ohm}$$

The magnetizing inductance obtained from the reactive component of the no-load test was:

$$L_m = 0.31 \text{ H}$$

Locked-Rotor Test For the locked-rotor test, the rotor was prevented from rotating, resulting in a slip equal to one. The measured quantities were:

$$V_{sc} = 39.7 \text{ V} \quad I_{sc} = 4.863 \text{ A} \quad P_{sc} = 299.99 \text{ W}$$

The short-circuit impedance was calculated as:

$$Z_{sc} = V_{sc} / I_{sc} = 8.16 \text{ ohm}$$

The equivalent series resistance was:

$$R_{sc} = P_{sc} / (3 \times I_{sc}^2) = 4.23 \text{ ohm}$$

Therefore, the rotor resistance referred to the stator was:

$$R_r = R_{sc} - R_s = 1.92 \text{ ohm}$$

The short-circuit reactance was approximately 6.95 ohm.

Assuming equal stator and rotor leakage inductances, the resulting values were:

$$L_{ls} = L_{lr} = 11.06 \text{ mH}$$

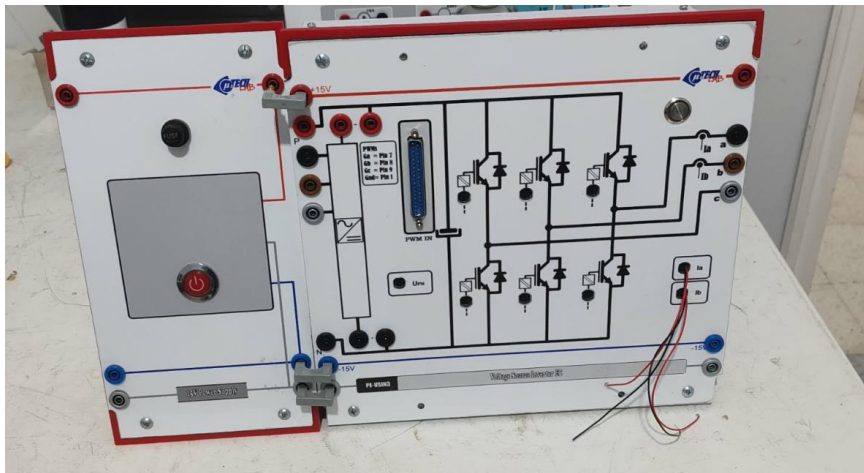
The final identified parameters are summarized below

Parameter	symbol	Measured value
Stator resistance	R_s	2.3
Iron loss resistance	R_f	1090.51
Magnetizing inductance	L_m	0.31
Referred rotor resistance	R_r	1.92
Stator leakage inductance	L_s	11.06 e-3
Rotor leakage inductance	L_r	11.06 e-3

parameters were subsequently used in the motor model and in the development of the control algorithms.

2.3 Three-Phase Inverter

After identifying the motor parameters, the three-phase inverter was prepared for the experimental implementation. The inverter is based on a two-level three-phase power stage consisting of six power switches and their associated gate-driver circuits. The board also integrates DC-link capacitors, current and voltage measurement circuits, and protection functions.

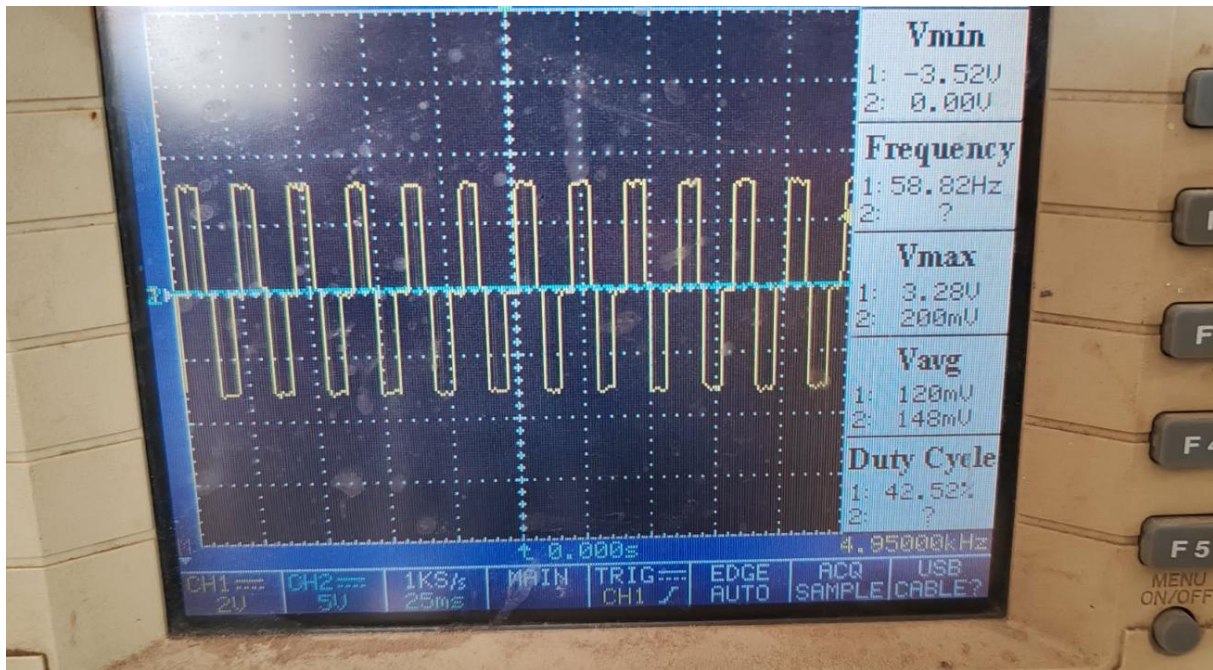


The inverter was supplied from a variable DC source available in the laboratory. Before connecting the motor, the power stage and gate-driver circuits were individually tested. The

commissioning process included the verification of the switching signals, protection functions, current measurements, and control interfaces.

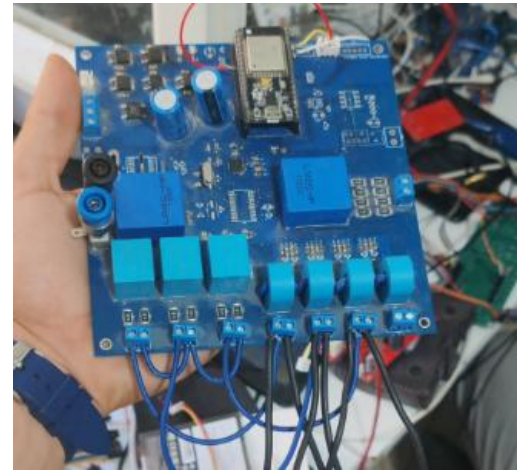
The overvoltage protection was adjusted to approximately **560 V**, while the thermal protection threshold was set to approximately **60 degrees C**.

After the hardware commissioning, the gate signals were verified using an oscilloscope. This confirmed the correct operation of the switching stage and the interface between the digital controller and the inverter.



2.4 Electrical Measurement System

A dedicated three-phase measurement unit was used to acquire the electrical quantities required during the experiments. The system includes current and voltage sensors followed by signal-conditioning and instrumentation circuits. The measured signals are processed by microcontrollers, allowing the calculation and visualization of quantities such as phase voltage, phase current, active power, and power factor. An ESP32-based communication interface was also used to transmit the measurement data through Wi-Fi to a monitoring dashboard.



Before being used in the experiments, the measurement system was calibrated using a multimeter and an oscilloscope to verify the correspondence between the measured electrical quantities and the signals provided to the acquisition system.

2.5 STM32-Based Control Platform

The digital control system was implemented using an STM32G4RE7 microcontroller. STM32CubeMX and STM32CubeIDE were used for configuring and programming the microcontroller, while an ST-LINK interface was used for programming and debugging. The STM32 was responsible for generating the PWM signals required by the inverter and executing the control algorithms in real time. Timer peripherals and interrupts were configured to provide synchronized switching and deterministic execution. The control development was also integrated with PLECS and Embedded Coder.

Two models were used:

One for simulation and one for embedded implementation. The control algorithm could therefore be tested in simulation before generating the C code and deploying it to the STM32. The general development process was:

Control algorithm -> PLECS simulation -> C code generation -> STM32 -> Real inverter

Establishing the communication between PLECS and the STM32 required additional configuration and testing.

Once this interface was operational, parameters and control variables could be exchanged between the simulation environment and the real embedded system.

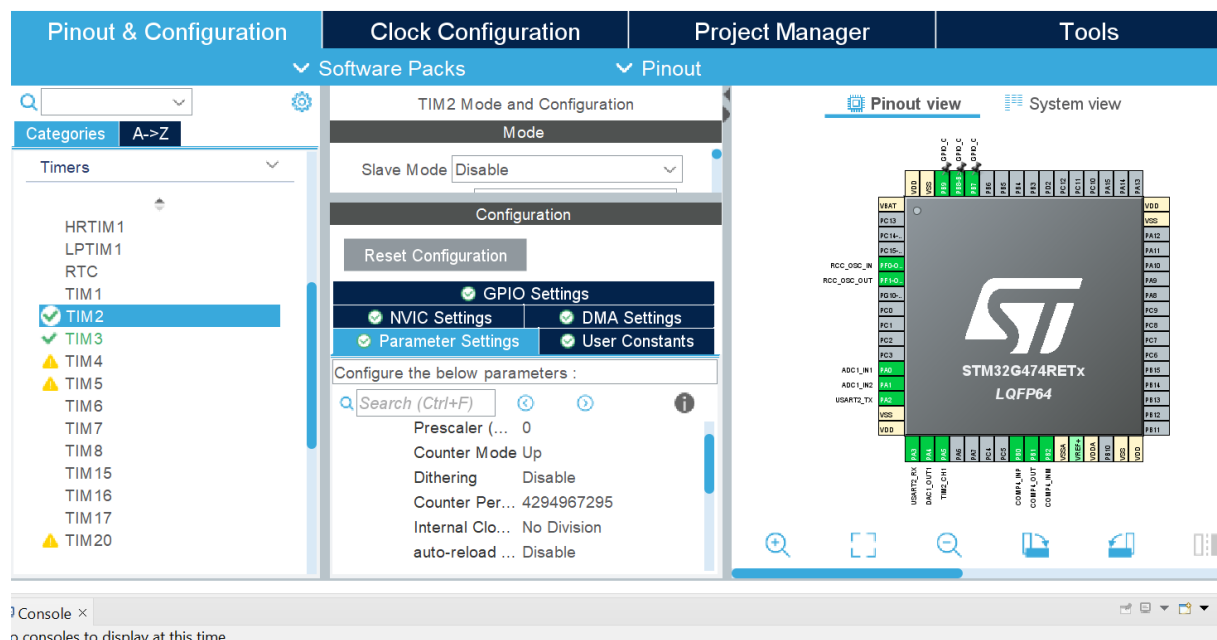
Experimental Platform and Preliminary Development

2.9 Digital PWM Generation and Preliminary Validation

After commissioning the inverter, measurement system, and STM32-based control platform, the next step was to investigate PWM generation for experimental implementation of the motor control strategies.

2.9.1 Basic PWM Generation

Using STM32 The first tests were carried out directly on the STM32G4RE7 using STM32CubeMX and STM32CubeIDE. PWM signals were generated using the internal timers of the microcontroller. The switching frequency and duty cycle were configured through the timer parameters. The generated signals were observed using an oscilloscope to verify their frequency, duty cycle, and switching behavior.



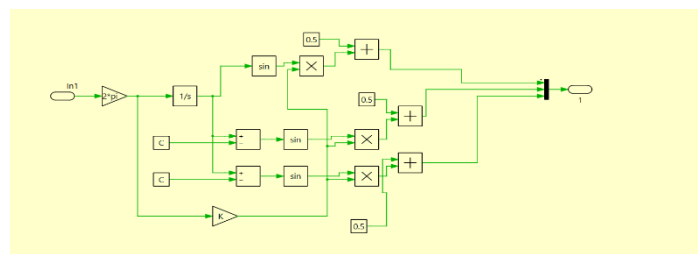
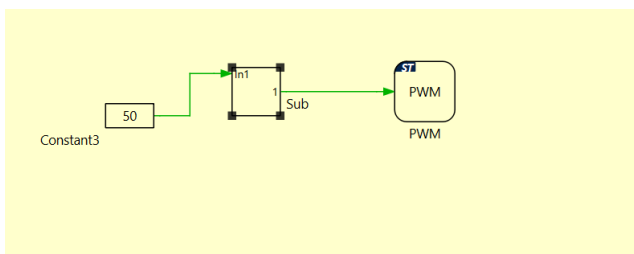
2.9.2 Sinusoidal PWM

Generation Sinusoidal pulse width modulation (SPWM) was then implemented to generate three-phase PWM signals. The three reference signals were generated with a phase displacement of 120 degrees. The switching frequency and modulation magnitude were varied experimentally in order to observe their effects on the generated signals.



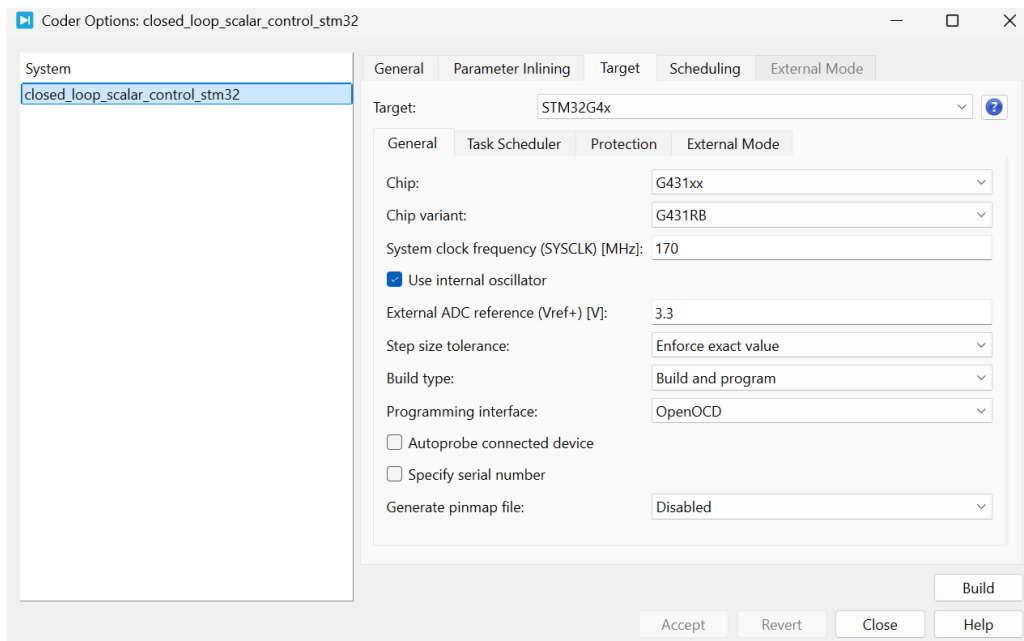
2.9.3 PLECS Implementation

The PWM generation was also implemented in PLECS. The model was first tested in simulation and then prepared for embedded implementation using Embedded Coder. The generated code was deployed on the STM32 controller and tested on the real experimental platform.



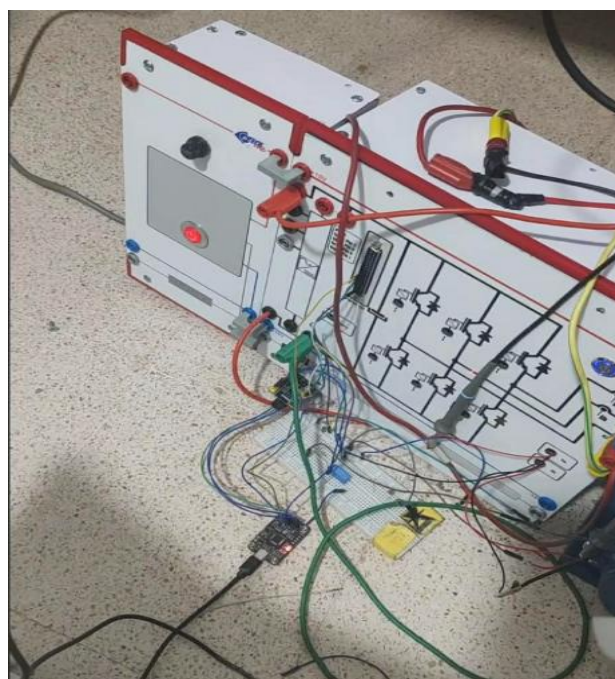
2.9.4 Real-Time Communication and External Mode

A communication link between PLECS and the STM32 controller was established to allow real-time interaction with the embedded system. PLECS External Mode was used to monitor the system and modify selected parameters during operation. This required configuring the communication between the software environment and the microcontroller.



2.9.5 Preliminary Experimental Validation

The PWM generation was finally tested on the real inverter. The three-phase PWM outputs were observed using an oscilloscope to verify the switching frequency, duty cycle, switching sequence, and phase displacement. LEDs and jumpers available on the control board were also used during the preliminary tests to verify the switching states and the phase relationships between the three inverter legs.



Space vector modulation (SVM) was also implemented and tested as part of the preliminary modulation study. Its behavior was first verified in simulation and then prepared for implementation on the STM32 platform.

2.10 Conclusion

The preliminary development successfully established the complete experimental chain from digital PWM generation to real-time execution on the STM32-controlled inverter. The inverter, measurement system, communication interface, and modulation algorithms were integrated and experimentally validated, providing the necessary foundation for the implementation of the motor control strategies presented in the following chapters.

Chapter 3

Implementation of Scalar Control and Closed-Loop Speed Regulation

3.1 Introduction

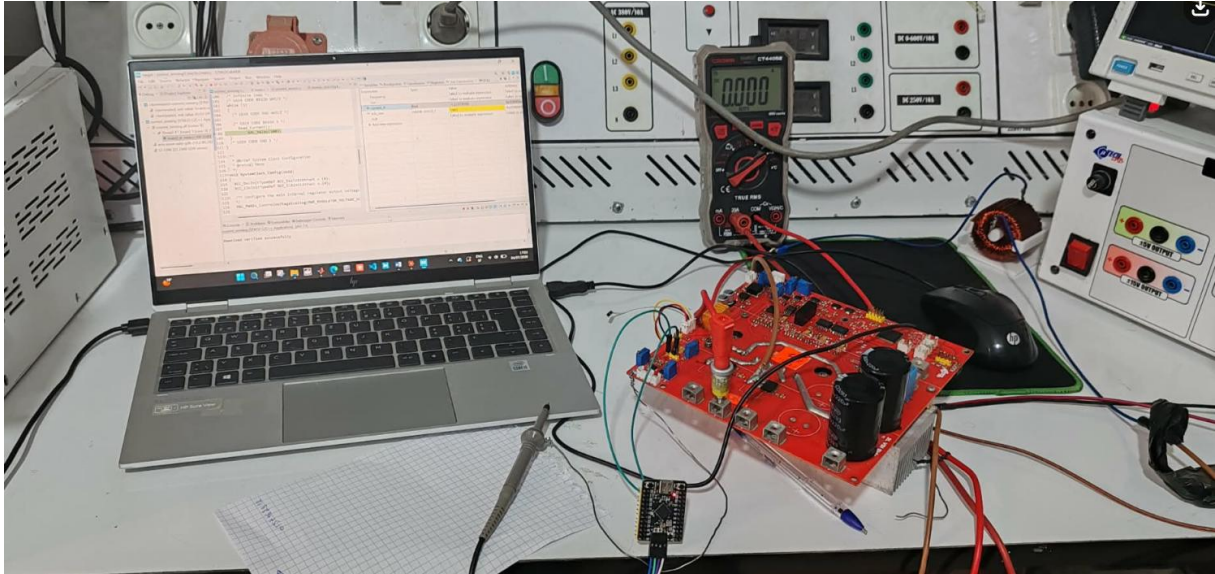
After completing the commissioning of the experimental platform and validating the PWM generation, the next stage of the internship focused on the implementation of scalar control for the induction motor.

The objective was first to obtain a reliable open-loop V/f control and then to extend the system to closed-loop speed control by introducing a real-time speed measurement. Unlike the previous stages, this part required the simultaneous operation of several time-critical functions on the STM32 microcontroller, including PWM generation, motor-control calculations, speed measurement, timer input capture, and interrupt handling. The implementation was therefore carried out progressively, starting with the validation of the final inverter and the speed-measurement system, followed by open-loop scalar control and finally closed-loop speed regulation.

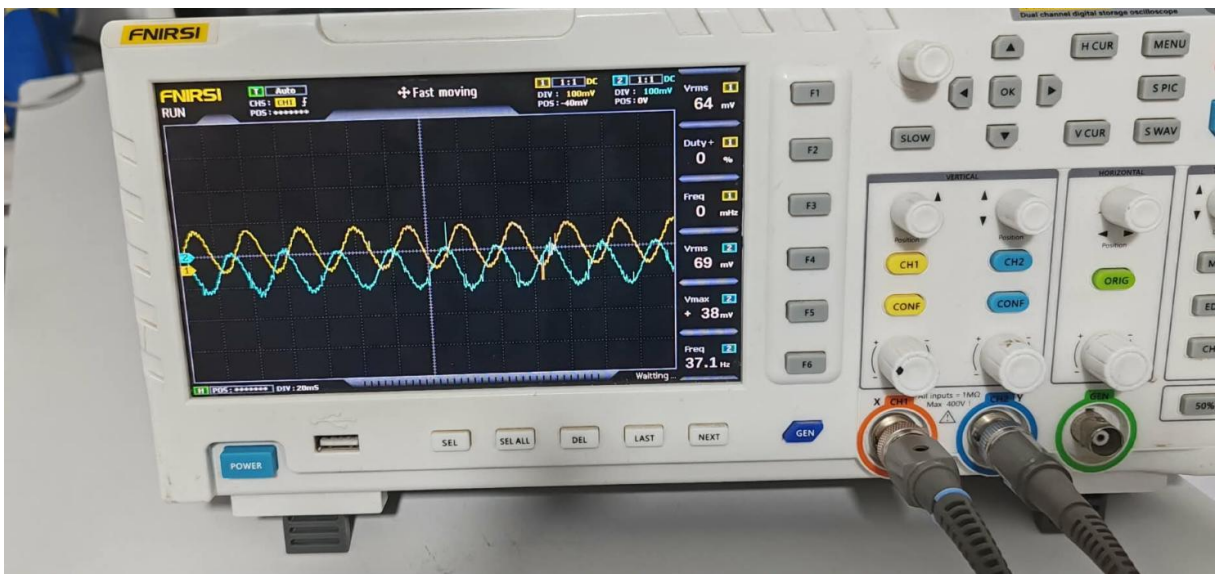
3.2 Final Inverter Commissioning and Power Stage

Validation Before applying the scalar-control algorithm to the motor, a final inverter was completely tested from the power stage to the control and measurement circuits. This step was necessary to ensure that the hardware could operate reliably under the conditions required for the experimental tests. The inverter was supplied from a DC source with a DC-link voltage of approximately 550 V.

Before connecting the motor, the different parts of the power stage and measurement system were checked separately. Particular attention was given to the voltage measurement, current measurement, temperature monitoring, and protection circuits.

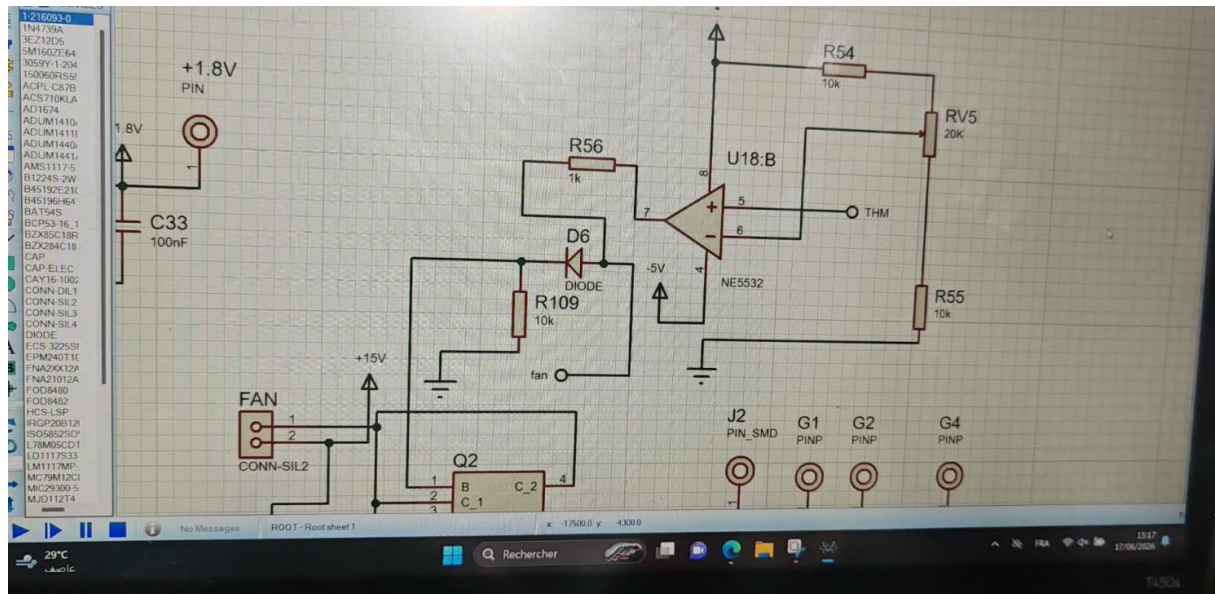


The current measurement system was tested by injecting known current values and comparing the sensor output with the expected signal. This made it possible to verify the complete measurement chain and determine whether the signal levels were suitable for acquisition by the STM32.

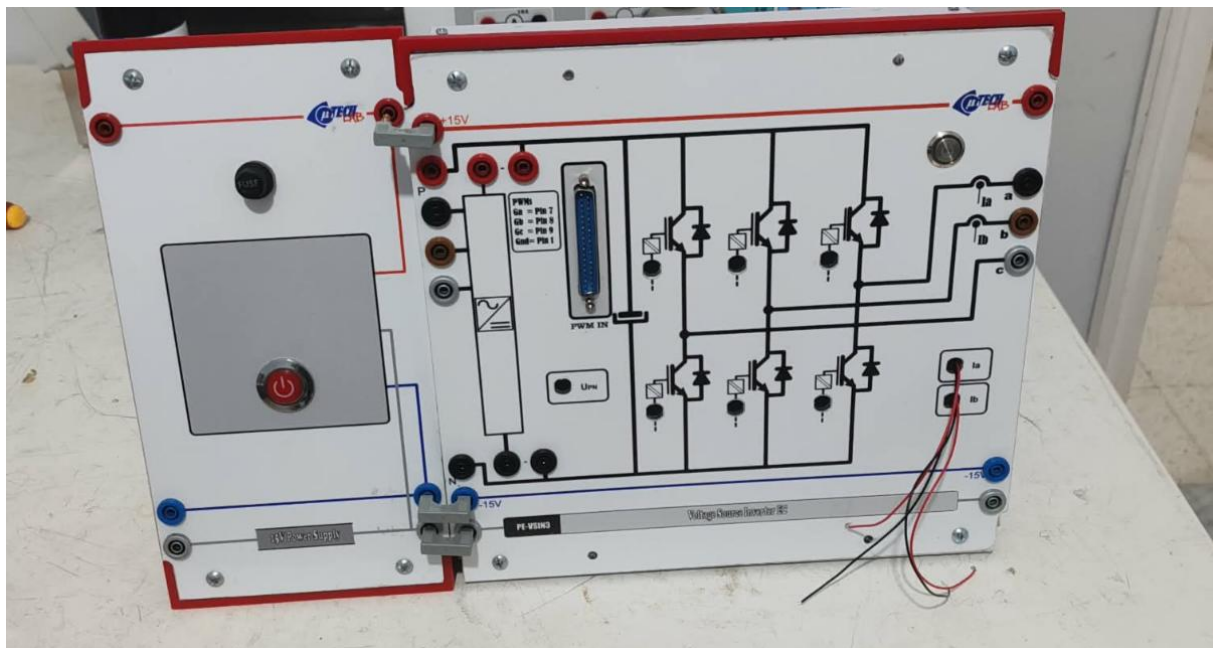


The voltage and temperature measurement circuits were also tested under controlled conditions. The corresponding signals were observed using laboratory instrumentation in order to verify their behavior and confirm that the measured values were within the expected ranges. A significant part of this work consisted of studying the available circuit documentation and component datasheets. Several missing components had to be soldered on

the PCB, and the instrumentation circuits had to be analyzed in order to determine the appropriate component values and signal levels. The required calculations were performed for the signal-conditioning circuits, particularly for the resistor networks used to adapt the sensor signals to the voltage range acceptable by the microcontroller.



This commissioning stage required approximately three weeks of testing, troubleshooting, measurements, and modifications before obtaining a reliable inverter that could be safely used for the motor experiments. Due to confidentiality restrictions, the detailed PCB layout, circuit references, and internal design documentation of the inverter cannot be included in this report.

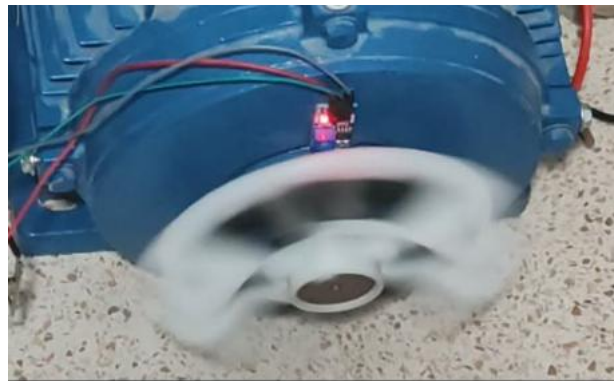


3.3 Development of the Motor Speed Measurement System

The implementation of *closed-loop scalar* control required a reliable measurement of the mechanical speed of the induction motor. For this purpose, an infrared sensor was selected as a simple and cost-effective solution.

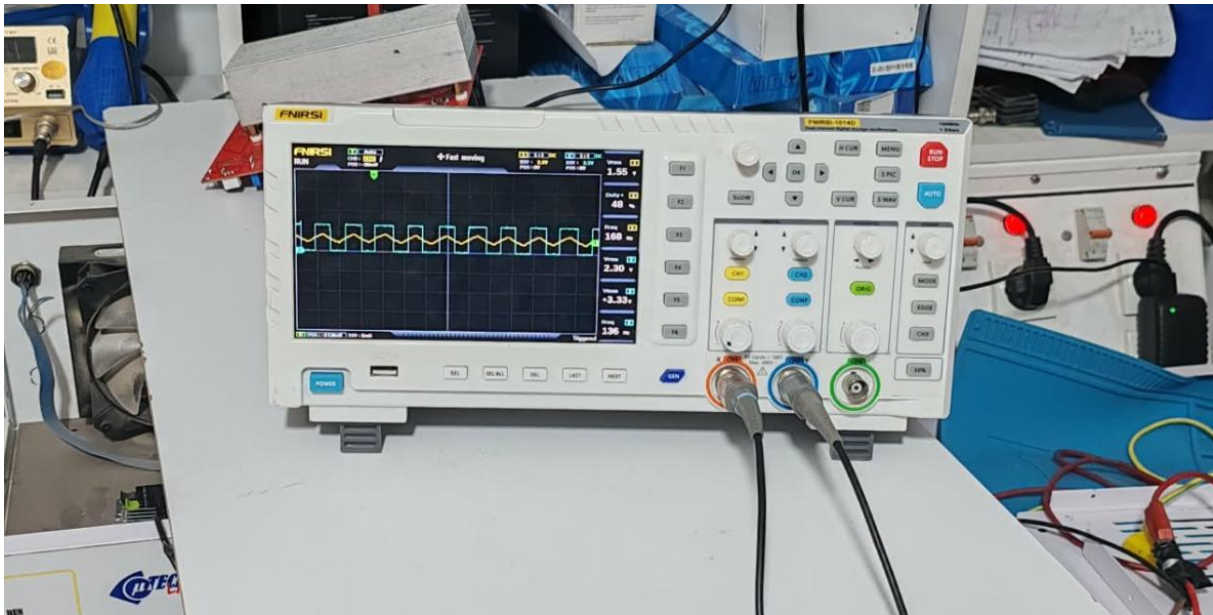


A mechanical disk containing six holes was designed within the laboratory and fixed to the motor shaft. The infrared sensor was mounted on the motor frame so that the passage of each hole could be detected during rotation. For one complete revolution, six holes pass in front of the sensor. Therefore, six rising edges of the sensor signal theoretically correspond to one mechanical revolution of the motor shaft.



The first experimental tests revealed that the optical detection was sensitive to the mechanical condition of the disk. The disk had to be cleaned and adjusted carefully to obtain a more stable detection signal. Even after improving the mechanical arrangement, the electrical signal contained a significant amount of noise. An **RC filter** was therefore introduced to attenuate the high-frequency disturbances and obtain a smoother signal.

However, filtering the signal was not sufficient to directly provide reliable digital transitions. A suitable threshold was required to distinguish the detected holes from the remaining signal variations. The internal comparator of the STM32 was therefore used to convert the filtered sensor signal into a clean digital signal suitable for edge detection.



3.4 STM32 Timer Input Capture for Speed Measurement

The first approach considered for detecting the sensor signal was a direct digital reading using the STM32. However, at the operating speed of the induction motor, this approach was not sufficiently reliable because the signal transitions occurred too rapidly for simple software polling. The STM32 timer *input capture mode* was therefore used. With input capture, each rising edge generated by the conditioned sensor signal is detected directly by the timer hardware. The timer records the corresponding counter value, allowing the time interval between two consecutive rising edges to be determined accurately. The frequency of the sensor signal was then calculated from this measured time interval.

Since six rising edges correspond to one mechanical revolution, the mechanical speed could be obtained from the measured signal frequency.

If T_{edge} represents the time between two consecutive rising edges, the motor speed can be expressed as:

$$\text{Speed (rpm)} = 60 / (6 \times T_{edge})$$

This method avoided the need for continuous software polling and made use of the dedicated timing capabilities of the STM32.

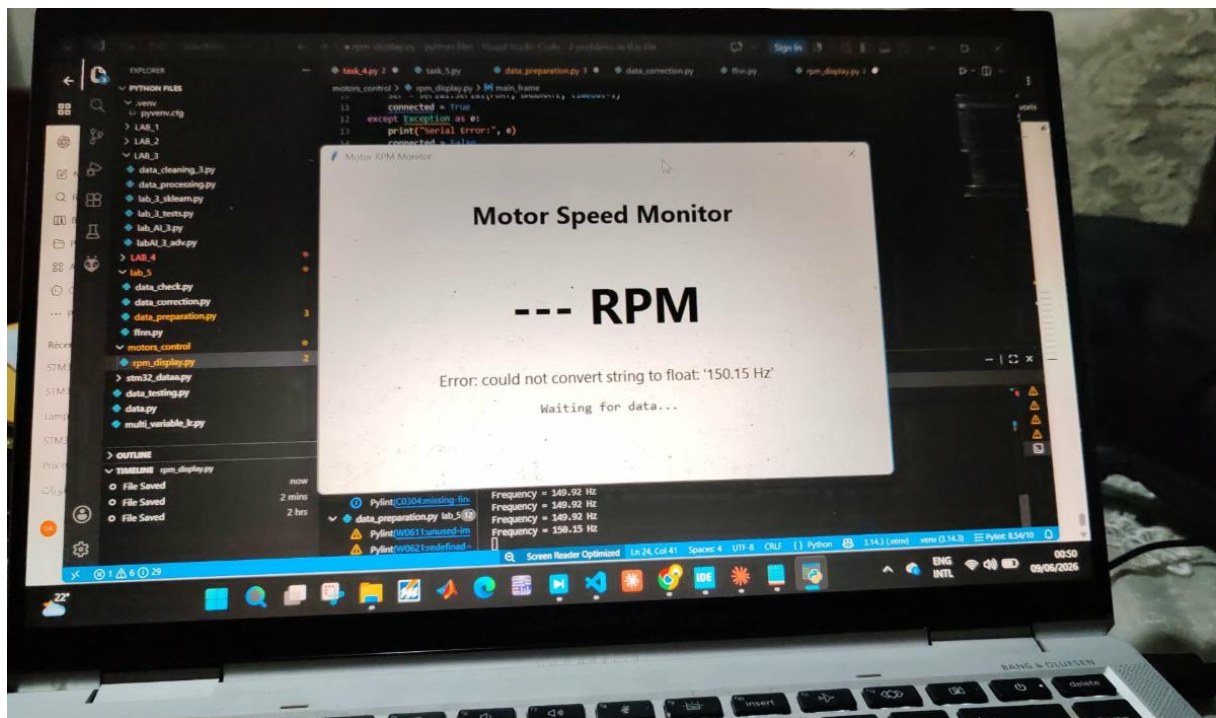
The complete frequency and speed calculation was implemented in STM32CubeIDE and tested directly with the running induction motor. The resulting speed measurement was successfully obtained in real time.

3.5 Experimental Validation of the Speed Measurement

The speed-measurement system was progressively validated by comparing the behavior of the sensor signal, the filtered signal, and the digital signal generated by the STM32 comparator.

The signal-conditioning stage was particularly important because the quality of the detected rising edges directly affected the accuracy of the measured period. The combination of mechanical adjustment, RC filtering, and comparator thresholding resulted in a sufficiently clean signal for the input-capture system.

The timer input capture was then used to measure the time between successive edges during motor operation. The measured period was converted into frequency and subsequently into mechanical speed. This approach provided a continuous speed measurement that could later be used as the feedback variable for the closed-loop controller.



3.6 Speed Measurement Implementation in PLECS

In parallel with the embedded implementation, the speed-measurement function was also developed and tested in PLECS. MATLAB and PLECS were used throughout the development of the control system to validate the mathematical relationships, simulate the control algorithms, and observe the expected behavior before transferring the functions to the STM32. The PLECS implementation provided a useful environment for testing the speed-

measurement algorithm and understanding its behavior before integrating it with the real-time embedded system.

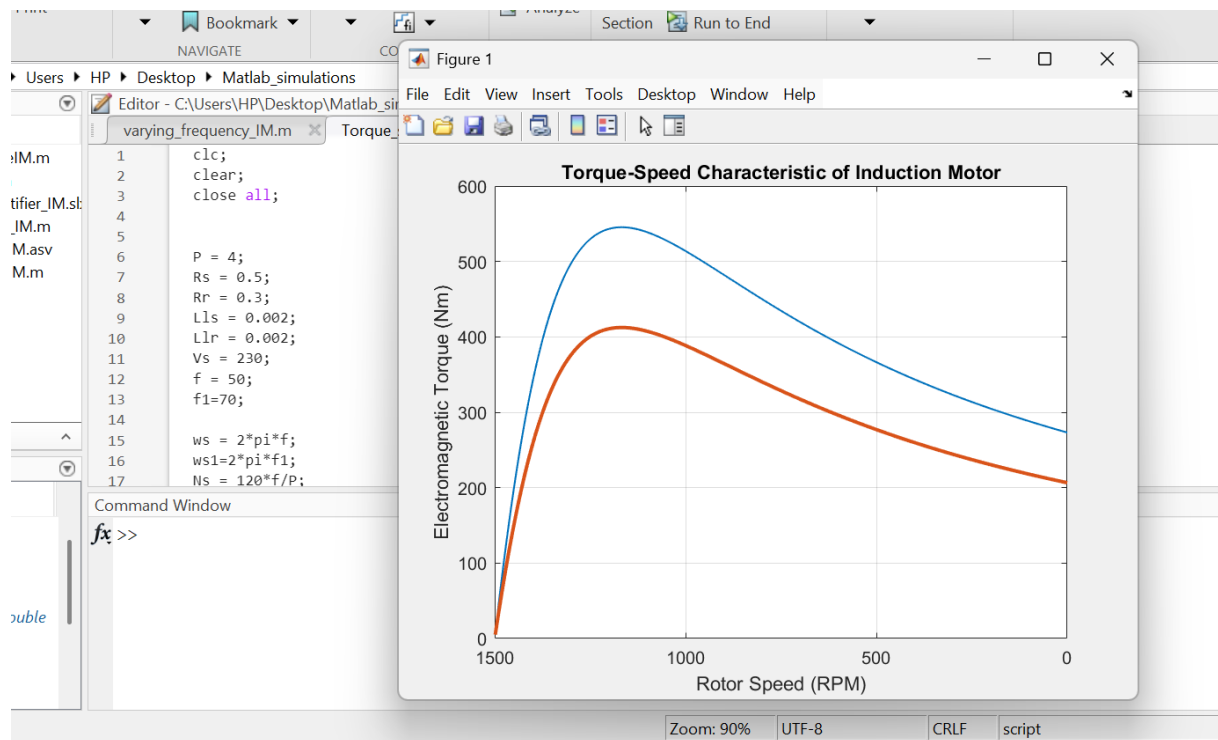


Figure: torque vs speed for induction machine (voltage variation)

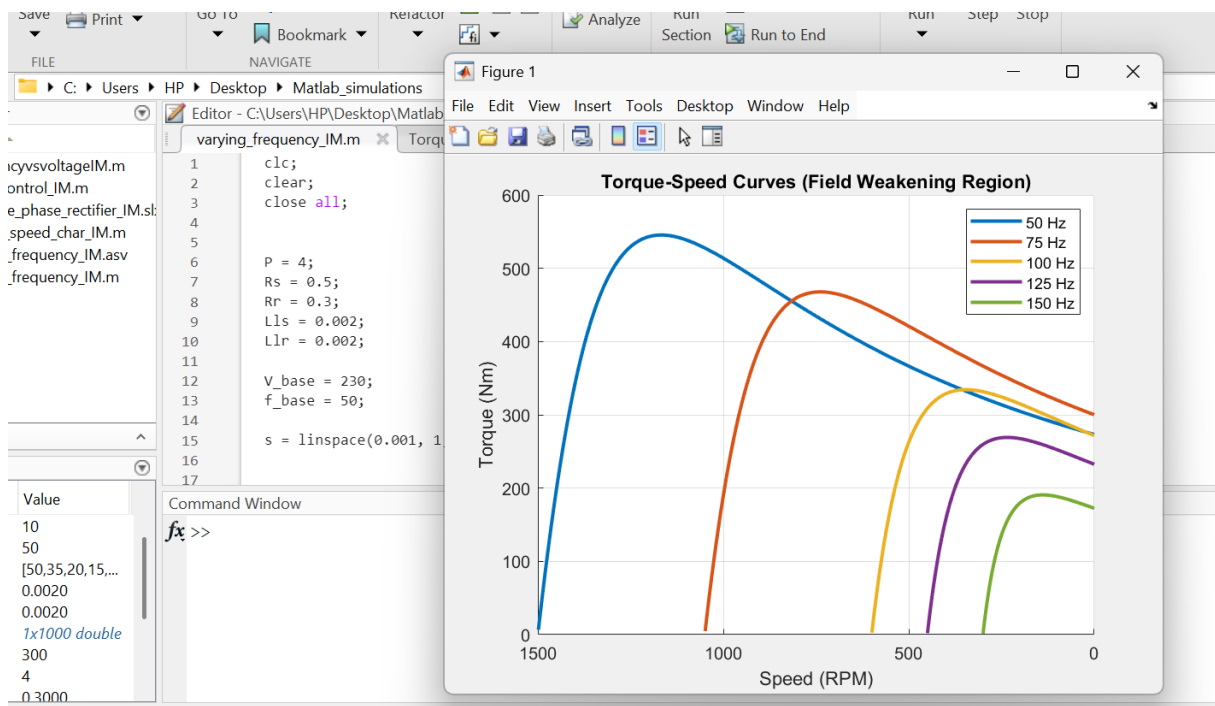
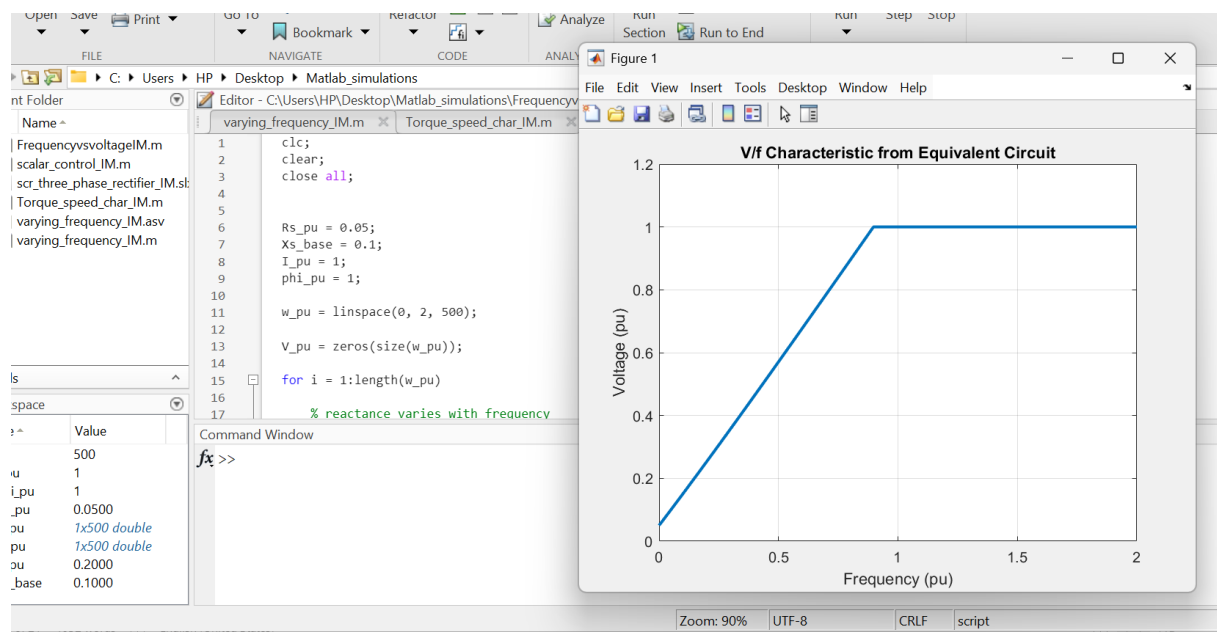


Figure: variation of the frequency in induction motor

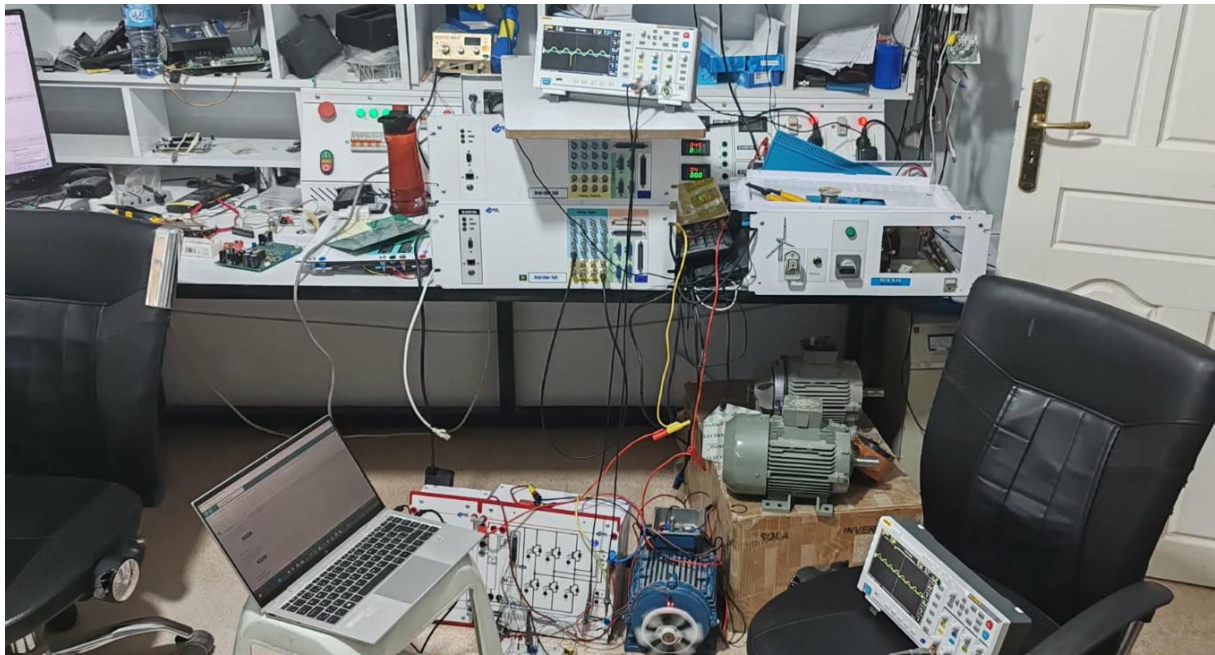
3.7 Open-Loop Scalar Control

Once the PWM generation and speed-measurement functions had been validated, the first motor-control stage consisted of implementing open-loop scalar control. The implemented strategy was based on the V/f principle. The stator voltage and electrical frequency were varied together in order to maintain an approximately constant voltage-to-frequency ratio over the normal operating range. In the low-frequency region, the voltage drop across the stator resistance becomes more significant relative to the applied voltage. A low-frequency voltage boost is therefore commonly introduced to compensate for this effect. In the present implementation, this low-frequency boost was neglected, and the control was based on the main V/f relationship.

The V/f characteristic and the corresponding scalar-control algorithm were first developed and tested in MATLAB and PLECS.

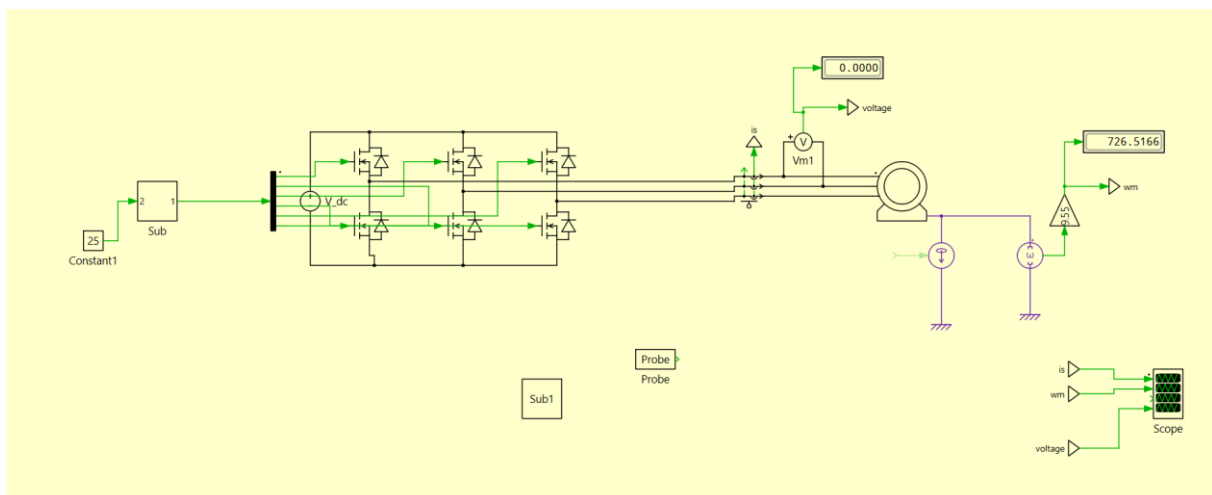


The simulation allowed the relationship between the frequency reference, voltage reference, and modulation signals to be verified before experimental implementation. The scalar-control model was then implemented on the STM32 and connected to the real inverter and induction motor. The open-loop scalar control was successfully operated without using speed feedback. This first experimental result confirmed that the PWM generation, inverter power stage, and V/f control algorithm were correctly integrated.



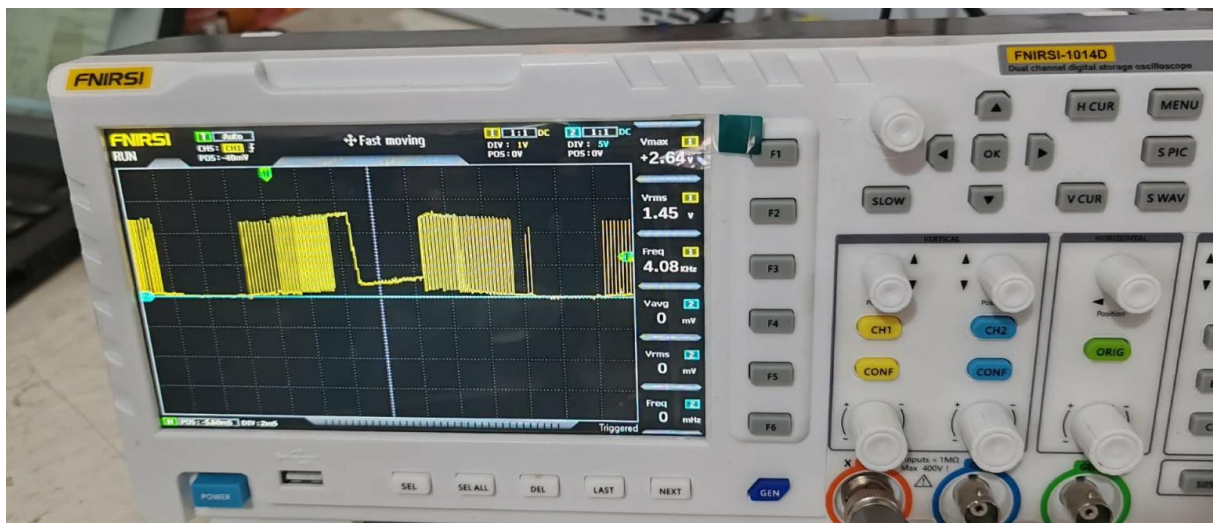
3.8 PLECS External Mode and Experimental Scalar Control

PLECS External Mode was used during the development of the scalar-control system to connect the simulation environment with the embedded controller. This allowed selected parameters to be modified during operation and made it possible to observe the effect of parameter changes on the real motor. The use of PLECS at this stage was particularly useful because it demonstrated the transition between the simulated control model and the real experimental system. The behavior predicted in simulation could be tested directly on the inverter and induction motor.



3.9 Integration of Scalar Control and Speed Measurement

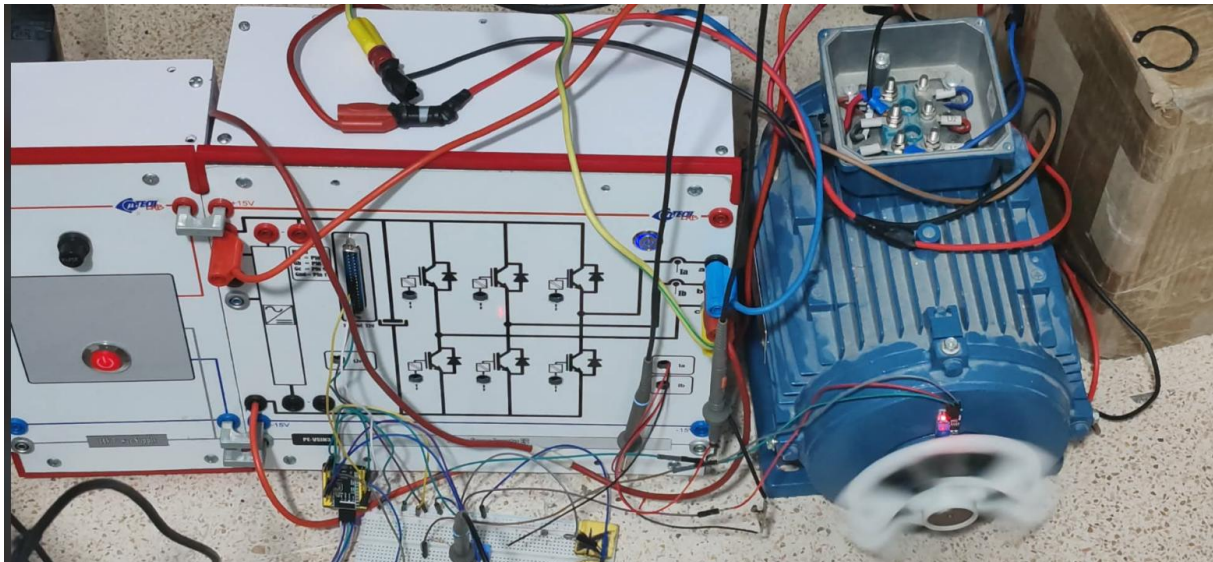
Interrupt Management and Real-Time Synchronization During the first attempts to combine scalar control and speed measurement, the motor did not behave correctly. The problem was not related to the scalar-control algorithm itself or to the speed sensor, but to the interaction between the different interrupt routines. The PWM generation was particularly sensitive to timing variations because the switching signals had to be generated with precise timing. At the same time, the speed-measurement system required interrupts whenever a rising edge was detected by the input-capture peripheral. The interrupt **priorities** were therefore carefully configured. The PWM generation and time-critical control operations were given higher priority than the speed-measurement processing.



This modification significantly improved the synchronization of the different real-time tasks and eliminated the abnormal motor behavior observed during the initial combined implementation. This debugging stage was particularly important because it demonstrated the importance of **interrupt management** when several hardware functions must operate simultaneously on a microcontroller.

3.10 Final STM32 Implementation

After resolving the timer and interrupt conflicts, the scalar-control and speed-measurement functions were successfully integrated into a single STM32 application. The STM32 was responsible for generating the PWM signals, executing the scalar-control algorithm, capturing the speed-sensor edges, calculating the frequency and motor speed, and executing the speed-control algorithm. The measured speed was therefore obtained directly from the hardware through the *input-capture* peripheral and made available to the control algorithm in real time. At this stage, the implementation was moved fully to STM32CubeIDE in order to obtain direct control over the microcontroller hardware and its available timers, interrupts, and peripherals. PLECS remained an important tool for simulation and algorithm validation, but the final real-time implementation of the combined system was performed directly on the STM32.



3.11 Closed-Loop Scalar Control

After obtaining a reliable real-time speed measurement, the scalar-control system was extended to closed-loop operation. The measured motor speed was compared with a reference speed to generate the speed error.

This error was then processed by a PI controller whose output modified the frequency reference of the **V/f control**.

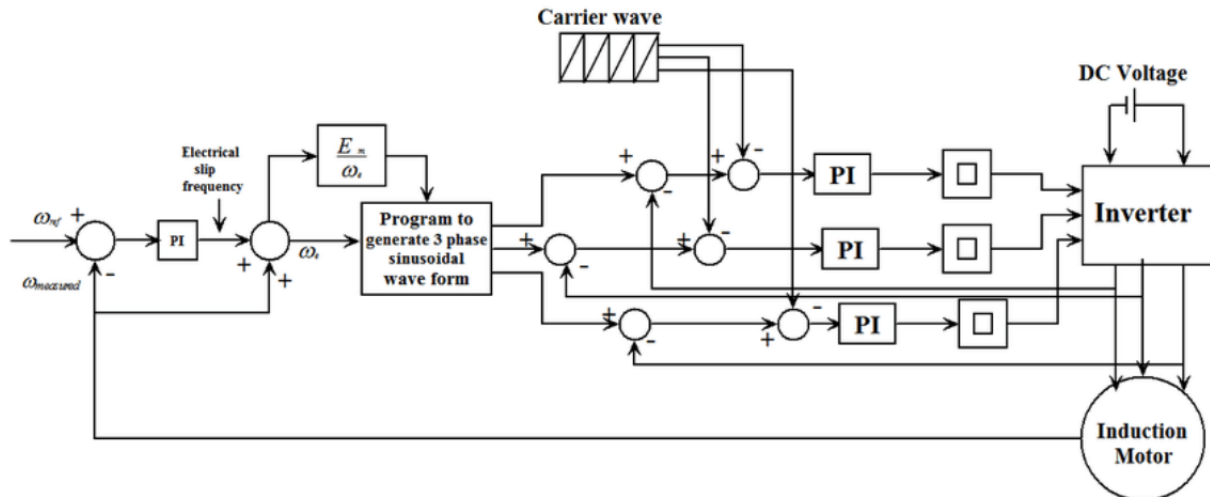
The controller parameters were tuned experimentally using a reference speed of 1200 rpm. The parameters used for the experimental tests were:

$$K_p = 0.56$$

$$K_i = 0.18$$

The output of the PI controller was used to modify the **frequency** command applied to the PWM generation stage.

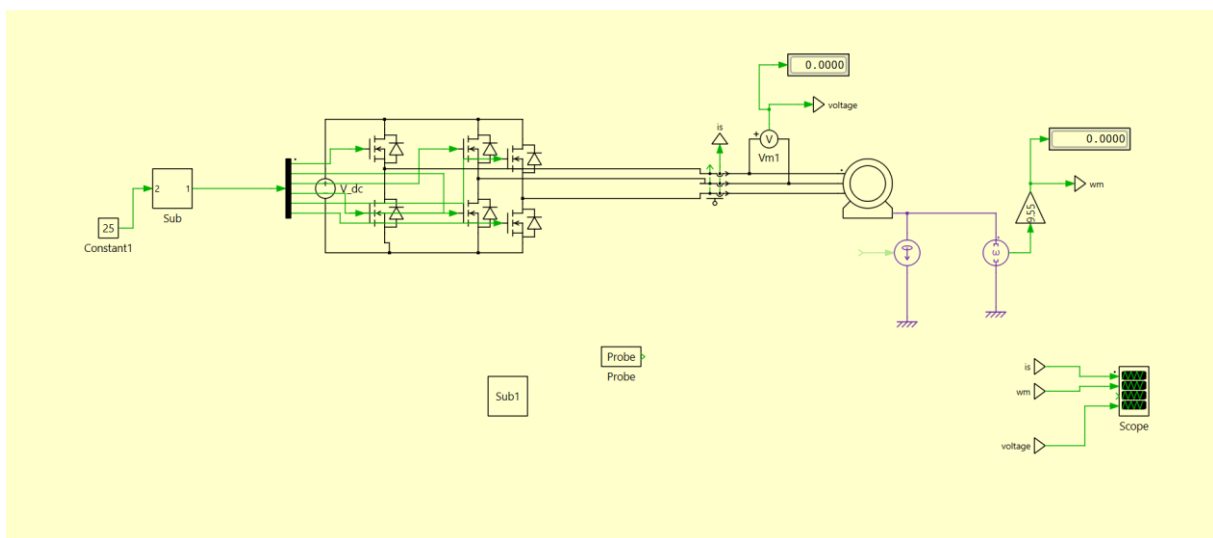
The timer responsible for generating the control frequency was therefore adjusted according to the controller output. The integral component required the knowledge of the time interval between successive control iterations. The embedded implementation therefore used the timing information available from the microcontroller to determine the effective sampling interval used in the integral calculation. This allowed the speed controller to operate according to the actual execution timing of the embedded system rather than relying only on an assumed fixed interval.



3.12 Experimental Closed-Loop Speed Regulation

The closed-loop scalar-control system was tested on the real induction motor using the commissioned inverter, the optical speed sensor, the signal-conditioning circuit, and the STM32 control system. A reference speed of 1200 rpm was used during the PI tuning process. Different reference conditions were also investigated to evaluate the response of the complete control system. Under the tested operating conditions, the motor reached a measured speed of approximately 1490 rpm.

3.14 MATLAB and PLECS Simulation Results MATLAB and PLECS were used throughout the development of the scalar-control system to validate the theoretical behavior and control algorithms. The V/f characteristic, modulation generation, speed measurement processing, and closed-loop control were studied in simulation before and during the experimental implementation. The simulation results were used as a reference for the experimental tests performed with the STM32 and the real inverter. This workflow made it possible to progressively move from the mathematical model to simulation and finally to real-time embedded control.



3.13 Overall Experimental Result

The scalar-control implementation resulted in a complete real-time control system operating on the STM32 microcontroller. The final system combined the power stage, voltage and current measurements, PWM generation, optical speed measurement, RC signal conditioning, comparator-based edge detection, timer input capture, frequency calculation, interrupt management, V/f control, and PI speed regulation. The main technical challenge was not the implementation of each individual function, but their simultaneous operation under the timing and hardware-resource constraints of the microcontroller. The resolution of these issues allowed the open-loop scalar control to be extended successfully to a closed-loop configuration with real-time speed feedback.

3.14 Conclusion

The scalar-control stage represented a major part of the experimental development carried out during the internship. Starting from the commissioning of the final high-voltage inverter, the work progressed through the development of a dedicated speed-measurement system, STM32 timer input capture, open-loop V/f control, real-time synchronization, interrupt management, and finally closed-loop speed regulation. The final implementation demonstrated the complete transition from simulation and algorithm development in MATLAB and PLECS to real-time control of a physical induction motor using the STM32 platform. This stage also provided significant practical experience in power-stage commissioning, instrumentation, embedded programming, real-time systems, timer management, interrupt prioritization, and closed-loop control.

Chapter 4

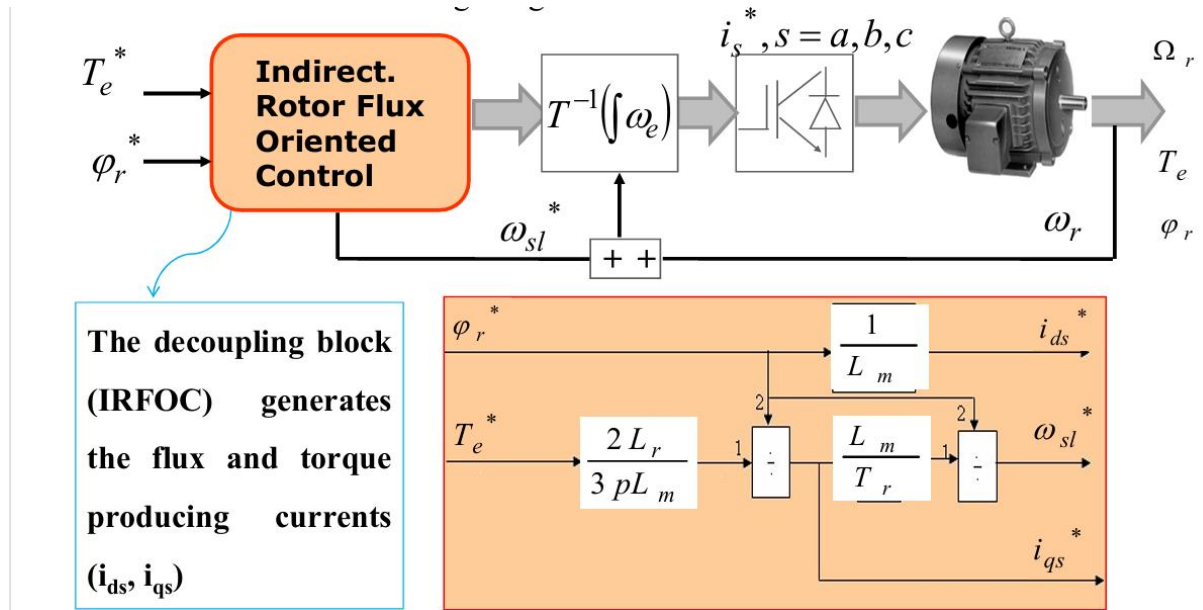
Implementation and Experimental Validation of Indirect Rotor Field-Oriented Control

4.1 Introduction

After successfully implementing open-loop and closed-loop scalar control, the next stage of the internship focused on the implementation of *Indirect Rotor Field-Oriented Control (IRFOC)*. The objective was to move from scalar control toward vector control by independently controlling the rotor flux and torque-producing components of the induction-motor current.



The IRFOC algorithm was first implemented and validated in PLECS using the identified motor parameters. Once the expected behavior had been confirmed in simulation, the different parts of the algorithm were progressively transferred to the STM32 controller and tested experimentally. A major part of this stage consisted of obtaining reliable real-time current measurements. The measured currents were acquired through the inverter ADCs, calibrated using offset and gain adjustments, and transmitted to a Python-based monitoring interface. This measurement system was then integrated with the Clarke and Park transformations and the hysteresis current controller.

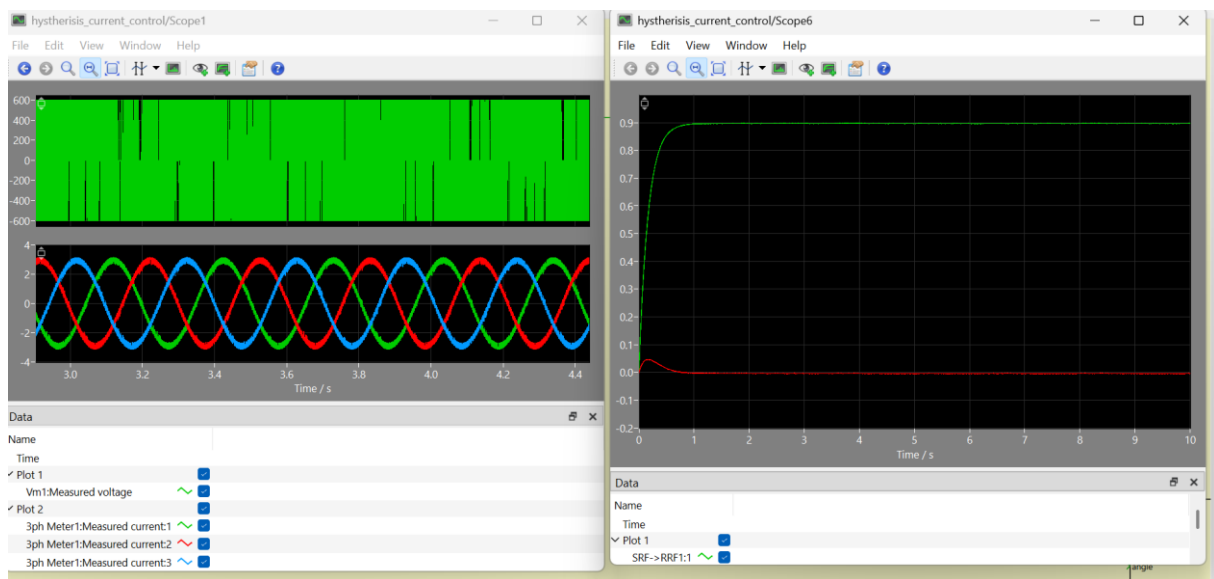


4.2 IRFOC Control Principle

The implemented control strategy was based on the *indirect rotor field-oriented control* principle. The main objective of IRFOC is to transform the three-phase stator currents into a rotating reference frame in which the current components can be associated with the rotor flux and electromagnetic torque. The measured three-phase currents are first transformed from the abc reference frame into the alpha-beta stationary reference frame using the Clarke transformation. The resulting alpha-beta currents are then transformed into the d-q rotating reference frame using the Park transformation. In the d-q reference frame, the d-axis current is mainly associated with rotor-flux production, while the q-axis current is associated with torque production. The reference flux and torque are therefore used to determine the required current references in the rotating reference frame. The reference currents are then transformed back into the three-phase reference frame and used by the current-control stage to generate the required inverter switching signals.

4.4 Simulation Validation

The simulation results obtained with PLECS provided the expected reference behavior for the experimental implementation. The identified motor parameters were introduced into the simulation model so that the simulated machine represented the real motor as closely as possible. The main signals observed during the simulation included the rotor flux, electromagnetic torque, d-axis and q-axis current references, stator current references, motor speed, and inverter switching signals. The simulation results were used as reference values for the experimental implementation. In particular, the simulated current waveforms and their amplitudes provided expected values that could later be compared with the currents measured directly on the real inverter.



4.5 Real-Time Current Acquisition

Using the STM32 After validating the control algorithm in simulation, the next challenge was to acquire the real motor currents accurately enough for the IRFOC algorithm. The inverter was equipped with current sensors, but the sensor signals first had to be acquired by the STM32 ADCs and calibrated before being used by the control algorithm. Two phase currents were directly measured. Since the three-phase motor currents form a balanced system, the third phase current can be calculated from the other two currents according to:

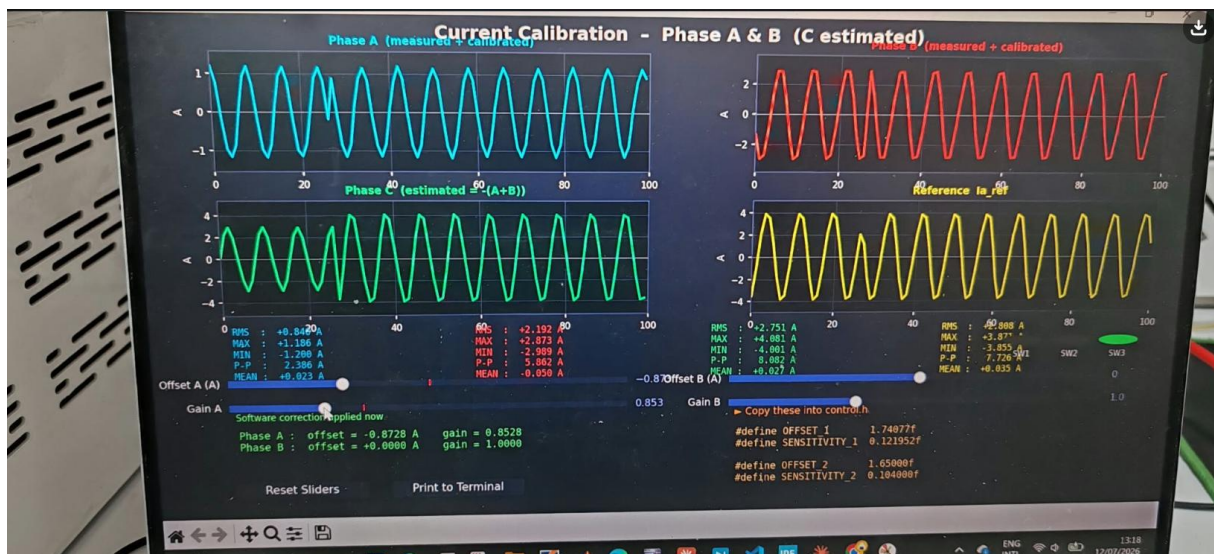
$$I_c = -(I_a + I_b)$$

This reduced the number of current sensors required while still allowing the three-phase current system to be reconstructed. The ADC configuration was therefore an important part of the implementation because the quality and timing of the acquired current signals directly affected the Clarke transformation, Park transformation, and hysteresis current controller.

4.6 Development of the Current Calibration System

A dedicated calibration and monitoring platform was developed to verify the current measurements before using them in the IRFOC algorithm. The measured ADC data were transmitted through the serial communication interface to a Python-based monitoring application. The application was used to visualize the acquired currents and provide adjustable calibration parameters. Separate offset and gain parameters were implemented for the two measured phase currents.

This monitoring system made it possible to compare the current values acquired by the STM32 with measurements obtained using laboratory instruments. The calibration process was divided into two main stages: *offset calibration and gain calibration*.



4.7 Current Sensor Offset Calibration

The first calibration stage consisted of correcting the offset of each current measurement channel. A known DC current condition was applied and the corresponding ADC output was observed through the Python monitoring interface. The offset parameter was progressively adjusted for the first current channel until the measured value corresponded to the expected reference. The same procedure was then repeated for the second current channel. This adjustment was particularly sensitive because small variations in the calibration parameters could produce noticeable changes in the measured current. The Python dashboard made this process considerably easier by allowing the effect of each parameter modification to be observed immediately.

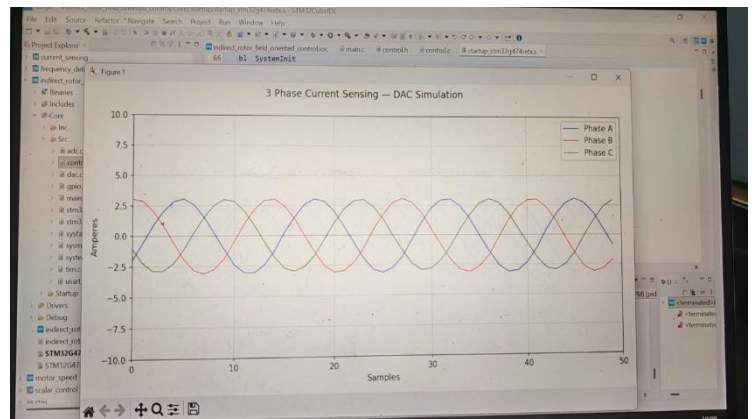
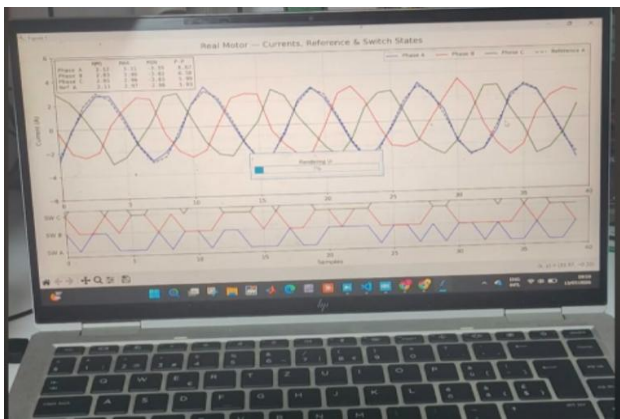
4.8 Current Sensor Gain Calibration

After correcting the offsets, an AC current was applied to the measurement system to calibrate the gain of each current channel. The gain parameters were adjusted until the amplitude measured by the STM32 corresponded to the actual injected current. The results were simultaneously monitored through the Python dashboard and laboratory oscilloscope.

This allowed the reconstructed current waveform from the STM32 to be compared directly with the physical current waveform. After the calibration process, the amplitude and waveform of the acquired current showed good agreement with the reference measurements.

4.9 Python-Based Real-Time Monitoring

The Python interface developed during this stage was used as a real-time monitoring and calibration tool. The acquired current signals were transmitted from the STM32 through the serial communication interface and plotted in real time. The interface allowed the offset and gain parameters of the two current channels to be modified during the calibration process. It was also used later during the IRFOC experiments to observe the measured currents and other control variables. This tool provided an additional method of validating the embedded measurements independently from the oscilloscope.



4.10 Implementation of Clarke and Park Transformations

Once the current acquisition was calibrated, the mathematical transformations required by the IRFOC algorithm were implemented in C. The Clarke transformation was used to convert the three-phase currents from the abc reference frame into the stationary alpha-beta reference frame. The Park transformation was then used to transform the alpha-beta currents into the rotating d-q reference frame. The inverse transformations were also implemented to convert the current references generated by the IRFOC algorithm back into the three-phase reference frame. The different transformations were implemented as independent C functions to simplify testing, debugging, and integration into the complete control algorithm.

4.11 Implementation of the IRFOC Algorithm

The complete IRFOC algorithm was then implemented on the STM32. The reference rotor flux and torque were used together with the identified motor parameters to determine the required d-axis and q-axis current references. The rotor slip speed was calculated from the reference torque and rotor flux. The slip speed was then combined with the measured rotor speed to obtain the electrical angular speed of the rotating reference frame. The resulting reference currents were transformed into the three-phase reference frame using the inverse

Park and Clarke transformations. The resulting phase-current references were then compared with the measured motor currents by the hysteresis current controller.

4.12 Hysteresis Current Control

The current control stage was based on hysteresis current control. The reference current of each phase was continuously compared with the corresponding measured current. According to the difference between the reference and measured currents, the inverter switching state was modified to force the actual current to remain within the defined hysteresis band around its reference. Before connecting the induction motor, the hysteresis current controller was first tested using an external RL load. This provided a safer environment for validating the current-control loop and checking the interaction between the ADC acquisition, current references, hysteresis controller, and inverter switching signals.

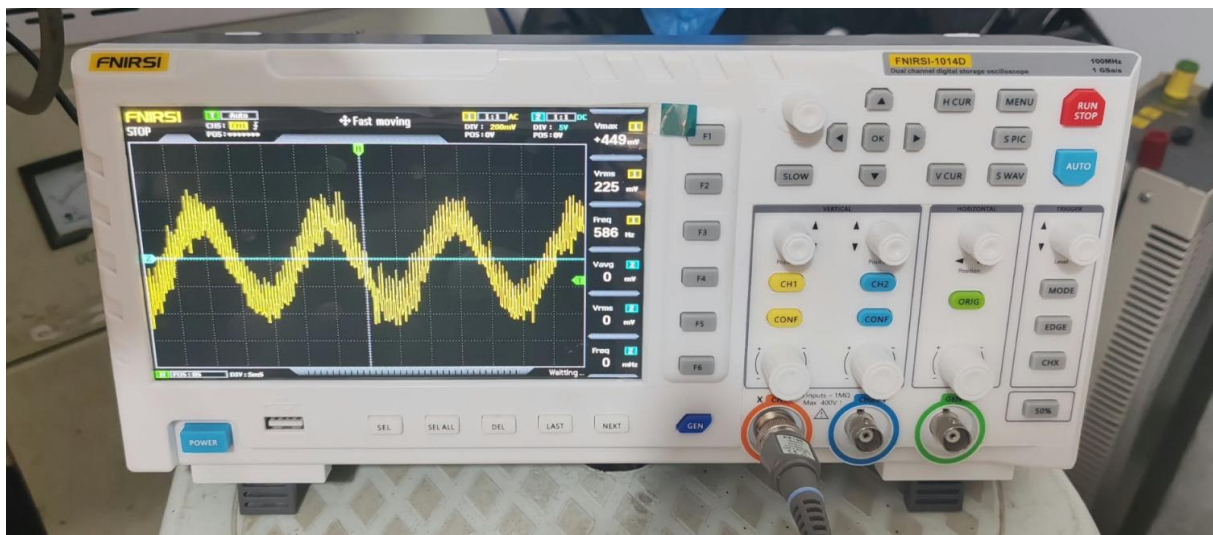


Figure: real time current generated

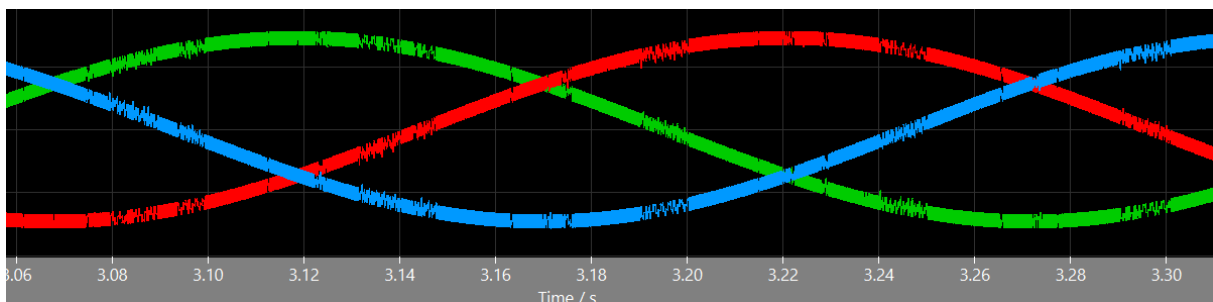
4.13 Experimental Implementation on the Induction Motor

After validating the hysteresis current controller with the external RL load, the complete IRFOC algorithm was connected to the real induction motor. The first motor tests revealed some differences between the expected and measured current signals. The remaining issues were mainly related to current-sensor calibration and the sign or inversion of some ADC measurement channels. The calibration parameters and ADC processing were therefore corrected before repeating the experiment. After these corrections, the IRFOC system operated successfully with the real induction motor. The measured currents, reference currents, and other control variables were observed simultaneously through the Python dashboard and laboratory oscilloscope.



4.14 Comparison Between Simulation and Experimental Results

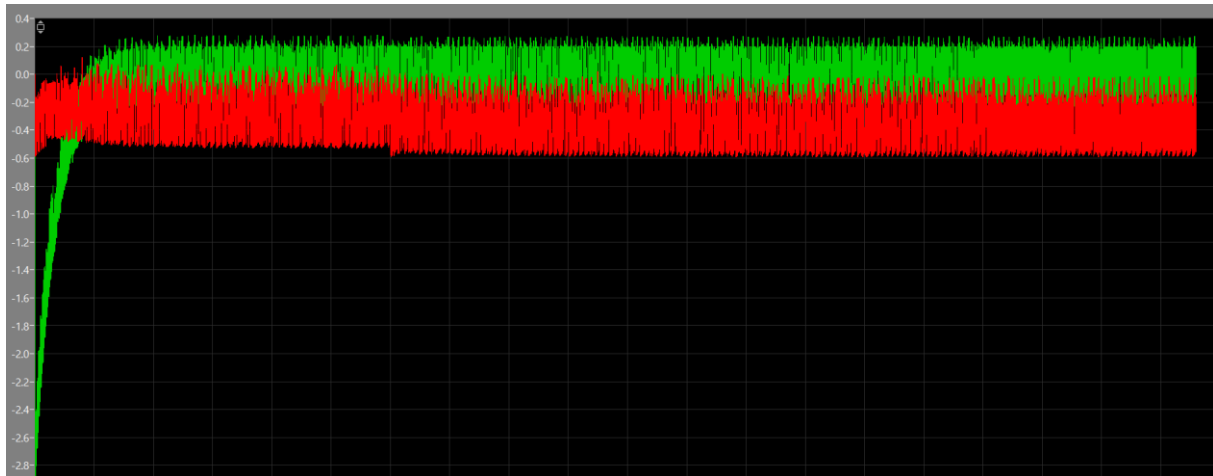
One of the main objectives of this stage was to compare the behavior predicted by the PLECS simulation with the behavior obtained from the real motor. The simulated and experimental current waveforms showed similar characteristics, including their general shape and amplitude evolution. The hysteresis current control was clearly observable in the experimental measurements, with the measured currents following their corresponding references within the expected control band. The agreement between the simulation and experimental results confirmed that the identified motor parameters and implemented control algorithm provided a representative model of the real system.



4.15 Experimental Load Test

After obtaining stable operation without an additional mechanical load, an additional mechanical load was applied to the motor to verify the behavior of the torque-control component of the IRFOC. The load was applied progressively while monitoring the phase currents, current references, and torque-related variables. As the mechanical load increased, the current required by the motor increased accordingly, while the torque-producing component remained regulated around its reference.

This experiment provided an additional validation of the decoupling principle of the IRFOC algorithm: *the control system was able to increase the torque-producing current component in response to the applied mechanical load while maintaining the flux-producing component according to its reference.*



4.16 Final Experimental Validation

The complete IRFOC implementation successfully combined the following functions: Three-phase current acquisition ADC calibration Offset and gain compensation Python-based real-time monitoring Clarke transformation Park transformation Inverse Park transformation Inverse Clarke transformation Rotor flux reference generation Torque reference generation Slip-speed calculation Hysteresis current control PWM and inverter switching Real-time operation on the STM32 The complete system was first validated in PLECS and then progressively transferred to the STM32 and the real induction motor. The experimental results obtained from the real system showed good agreement with the expected simulation behavior, particularly for the current waveforms and hysteresis current regulation.

4.17 Conclusion

The implementation of IRFOC represented the most advanced stage of the motor-control development carried out during the internship. The work progressed from simulation in MATLAB and PLECS to real-time implementation on the STM32. Particular attention was given to the calibration and synchronization of the current measurement system, since accurate and correctly timed current acquisition was essential for the Clarke and Park transformations and the hysteresis current controller. The successful integration of ADC acquisition, current calibration, coordinate transformations, rotor-flux-oriented control, slip-speed calculation, hysteresis current control, and inverter switching resulted in a complete experimental IRFOC system operating with the real induction motor. The comparison between simulation and experimental measurements provided an important validation of both the identified motor parameters and the implemented control algorithm. The experience also strengthened the practical skills acquired during the previous stages, particularly in embedded

programming, real-time control, ADC configuration, signal conditioning, debugging, and power-electronics experimentation.

Chapter 5

General Conclusion and Acknowledgements

5.1 General Conclusion

This internship at Microtech Lab provided an extensive practical experience in power electronics, electrical drives, embedded control, and experimental engineering. The work began with the characterization of the induction motor and the identification of its main electrical parameters. This first stage provided the parameters required for the subsequent simulations and control implementations. The experimental platform was then progressively prepared through the commissioning of the three-phase inverter, calibration of the voltage and current measurement systems, and configuration of the STM32-based control platform.

A significant part of the internship was dedicated to the development and validation of PWM generation. Different modulation approaches were studied and implemented using STM32 and PLECS, providing the necessary foundation for the subsequent motor-control strategies. The scalar-control stage then allowed the transition from basic PWM generation to complete real-time motor control. Open-loop V/f control was first implemented and validated before being extended to closed-loop speed control.

The development of the optical speed measurement system, signal conditioning, timer input capture, interrupt management, and PI speed regulation represented an important part of this work.

The final major stage consisted of implementing indirect rotor field-oriented control. The algorithm was first validated in PLECS using the identified motor parameters and was then progressively transferred to the STM32 controller. The implementation required the development of Clarke and Park transformations, real-time current acquisition, ADC calibration, current-sensor offset and gain compensation, rotor-flux and torque reference calculation, slip-speed calculation, and hysteresis current control.

The comparison between simulation and experimental results demonstrated the importance of combining theoretical modeling with practical measurements. Several difficulties encountered during the internship, particularly those related to PCB commissioning, ADC synchronization, timer resources, signal conditioning, and interrupt priorities, required iterative testing and debugging. These challenges provided valuable practical experience that could not be obtained from simulation alone.

The internship also provided experience with MATLAB, PLECS, STM32CubeMX, STM32CubeIDE, Embedded C, Python, oscilloscopes, laboratory power supplies, professional multimeters, current and voltage sensors, and real power-electronic hardware. Overall, the internship allowed the complete development process to be experienced, from theoretical modeling and simulation to embedded implementation and experimental validation on a real induction motor.

It also provided a better understanding of the relationship between power electronics, electrical machines, control theory, embedded systems, instrumentation, and real-time programming.

5.2 Professional and Technical Skills Acquired

This internship significantly strengthened my practical engineering skills. From a power-electronics perspective, I gained experience in three-phase inverter commissioning, power-stage testing, protection verification, current and voltage measurement, signal conditioning, and safe operation of a high-voltage DC link.

From an embedded-systems perspective, I developed practical experience with STM32 microcontrollers, timers, PWM generation, ADC acquisition, input capture, interrupts, interrupt priorities, serial communication, and real-time execution. From a control perspective, I implemented and experimentally validated scalar V/f control and indirect rotor field-oriented control. I also developed practical experience with coordinate transformations, current regulation, speed feedback, PI control, and real-time control implementation.

The internship also improved my debugging methodology. Several problems could not be solved simply by modifying the control algorithm and required investigation at the hardware, signal-conditioning, peripheral, and software levels. This experience reinforced the importance of systematic testing and measurement when working with real engineering systems.

5.3 Final Remarks

One of the most valuable aspects of this internship was the opportunity to work directly with real industrial and laboratory hardware rather than limiting the work to simulation. The progression from simulation to implementation provided a clearer understanding of the practical limitations of real systems. Sensor noise, ADC timing, signal offsets, hardware limitations, timer resources, interrupt priorities, and measurement errors all had to be considered in the final implementation. The experience gained during this internship constitutes an important step in my development as an electrical and power-electronics engineer.

It has strengthened both my theoretical understanding and my ability to implement, test, debug, and validate real-time electrical-drive systems.

5.4 Acknowledgements

I would like to express my sincere gratitude to **Microtech Lab** for giving me the opportunity to carry out this internship in a highly technical and stimulating environment. I would particularly like to thank Professors **mr A.khaldoun** and **mr B.metidji** for their support, guidance, and valuable technical advice throughout the internship. His support was especially important during the most challenging stages of the project, particularly during the debugging and real-time implementation of the motor-control system.

I would also like to sincerely thank the technical team and all the engineers, technicians, researchers, and staff members of **Microtech Lab** for their assistance, availability, and contribution to the successful completion of this work.

I am particularly grateful for the opportunity to work with the laboratory's experimental equipment and real power-electronics systems. The technical discussions, practical assistance, and experience shared by the members of the laboratory contributed significantly to my learning throughout the internship. Finally, I would like to thank everyone at Microtech Lab who contributed directly or indirectly to this experience and helped create a professional environment in which I was able to develop my technical and engineering skills.